

```
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```

```
% Matlab ver >= R2017b
```

```
%
```

```
% Carboplatin-inducd BM tox model
```

```
clear all
```

```
close all
```

```
%% MODEL PARAMETERS
```

```
% CARBO PK
```

```
QC = 0.06;% liter/hour
```

```
Vcen_Carbo = 0.26;% liter
```

```
Vper_Carbo = 0.30;% liter
```

```
Cl_carbo = 0.42;% liter/hour
```

```
kdamage = 0.017;% 1/hour
```

```
% BASELINE (molecule)
```

```
Neut0 = 1.32;
```

```
Mono0 = 0.16;
```

```
Ret0 = 235;
```

```
Plt0 = 967;
```

```
CMP0 = 214;
```

```
MEP0 = 156;
```

```
MPP0= 69;
```

```
lambda1 = 2;
```

```
lambda2 = 2;
```

```
RBC0 = 8161;
```

```
% CIRCULATING RATEs (1/h)
```

```
kcircNeut = 0.17;
```

```
kcircMono = 0.06;
```

```
kcircRBC = 0.0007;
kcircRet = kcircRBC*RBC0/Ret0;
kcircPlt = 0.01;
% MTTs (h)
MTTNeut = 61.5;
MTTMono = 62.2;
MTTRet = 75;
MTTPlt = 74;
% DRUG EFFECTS (1/uM)
SlopeMPP = 2.05;
SlopeCMP = 1.47;
SlopeMEP = 2.19;
deathTR = 2.8; %dimensionless
deathTP = 0.54; %dimensionless
% FEEDBACK (dimensionless)
gaMPP = 0.07;
gaT1T2 = 0.7;
gaMPPMatCMP = 0.6;
gaMPPMatMEP = 0.3;
alphaWBC = 0.5;
% PROLIFERATION RATES (1/h)
lambda3 = 1.8;
lambda4 = 1.8;
% SOURCE
lambda5 = 1.8;
```

FILE with ODE system

```
function dydt = ODERatModel_sharedCode(~,y,k,kPK)
```

```
% PARAMETER MAPPING
```

```
kcircNeut = k(1);
```

```
kcircMono = k(2);
```

```
kcircRet = k(3);
```

```
kcircPlt = k(4);
```

```
kcircRBC = k(5);
```

```
% BASELINE (molecule)
```

```
Neut0 = k(6);
```

```
Mono0 = k(7);
```

```
Ret0 = k(8);
```

```
Plt0 = k(9);
```

```
CMP0 = k(10);
```

```
MEP0 = k(11);
```

```
MPP0= k(12);
```

```
lambda1 = k(13);
```

```
%T10Ret = k(13);
```

```
lambda2 = k(14);
```

```
%T10Plt = k(14);
```

```
RBC0 = k(15);
```

```
% MTTs (h)
```

```
MTTNeut = k(16);
```

```
MTTMono = k(17);
```

```
MTTRet = k(18);
```

```
MTTPlt = k(19);
```

```
% DRUG EFFECTS (1/uM)
```

```
SlopeMPP = k(20);
```

```
SlopeCMP = k(21);
```

```
SlopeMEP = k(22);
```

```

deathTR = k(23);
deathTP = k(24);
% FEEDBACK
gaMPP = k(25);
gaT1T2 = k(26);
%gaDeDiffMPP = k(27);
gaMPPMatCMP = k(27);
gaMPPMatMEP = k(28);
alphaWBC = k(29);
% PROLIFERATION RATES (1/h)
%kprolCMP = k(31); % kprolCMP < ktrNeut + ktrMono
lambda3 = k(30);
%kprolMEP = k(32); % kprolMEP < ktrRet + ktrPit
lambda4 = k(31);
% SOURCE
lambda5 = k(32);

% CARBO PK
QC = kPK(1);
Vcen_Carbo = kPK(2);
Vper_Carbo = kPK(3);
Cl_carbo = kPK(4);
% EFFECT COMPARTMENT
kdamage = kPK(5);

ke_Carbo = Cl_carbo/Vcen_Carbo;
k12_Carbo = QC/Vcen_Carbo;

```